

**VELS INSTITUTE OF SCIENCE, TECHNOLOGY AND ADVANCED
STUDIES**

DEPARTMENT:INFORMATIO TECHNOLOGY

SEMESTER: I

TITLE OF THE COURSE: OBJECT ORIENTED SOFTWARE ENGINEERING

COURSE CODE: 24DMIT11

TIME: 03 Hours

Max. Marks: 100

SECTION – A (3 MARKS)

S. No	Questions	Unit	K-Level	CO	PO
1.	What is Software Engineering?	I	K1	CO2	PO1
2.	List the key challenges faced in Software Engineering?	I	K2	CO1	PO2
3.	What are the main objectives of Software Engineering?	I	K1	CO2	PO1
4.	Define the importance of Software Engineering?	I	K2	CO3	PO2
5.	What is the difference between Software Engineering and Programming?	I	K1	CO2	PO2
6.	What are the different phases in the Software Development Life Cycle	I	K1	CO1	PO2
7.	Define the role of a Software Engineer?	I	K2	CO3	PO2
8.	List the difference between a Software Engineer and a System Analyst?	I	K3	CO2	PO3
9.	Define a Software Project?	I	K2	CO2	PO1
10.	Define a Software Process?	I	K1	CO1	PO1
11.	What are the main objectives of a Software Process?	I	K1	CO2	PO2
12.	What is a Perspective Process Model?	I	K1	CO2	PO2
13.	What is the Waterfall Model in Software Process?	I	K1	CO3	PO1
14.	List out the Incremental Model in Software Process?	I	K3	CO2	PO2
15.	What is the V-Model in Software Process?	I	K2	CO1	PO1
16.	Define the Agile Process Model?	I	K3	CO2	PO2
17.	List out the Specialized Process Models in Software Engineering?	I	K4	CO2	PO1
18.	List out the key principles of Agile development?	I	K1	CO2	PO2
19.	What is Extreme Programming (XP) in Agile?	I	K2	CO3	PO3
20.	List out the key practices of Extreme Programming (XP)?	I	K2	CO3	PO2
21.	What is the purpose of requirement analysis in software engineering?	II	K3	CO2	PO3
22.	List the difference between functional and non-functional requirements?	II	K2	CO3	PO2
23.	Define "Requirement Specification"	II	K3	CO4	PO4
24.	Differentiate between "User Requirements" and "System Requirements."	II	K4	CO2	PO2

25.	Explain the role of a "Requirements Traceability Matrix" (RTM).	II	K3	CO2	PO1
26.	List out the Functional Requirements in a Software Requirement Specification (SRS)?	II	K2	CO2	PO3
27.	Define Software Requirement Specification (SRS)	II	K2	CO2	PO2
28.	What is a Formal System Specification?	II	K2	CO2	PO3
29.	List out the role of Stakeholders in Requirements Gathering?	II	K2	CO2	PO3
30.	List out the role of a Use Case in Requirements Specification?	II	K2	CO2	PO4
31.	Explain the role of Class Diagrams in UML.	II	K2	CO5	PO3
32.	What is an Interaction Diagram in UML?	II	K2	CO2	PO3
33.	Define a State Transition Diagram?	II	K2	CO3	PO2
34.	What is the difference between Sequence Diagrams and Collaboration Diagrams in UML?	II	K2	CO2	PO3
35.	What is a Level 0 DFD and how is it used?	II	K2	CO2	PO3
36.	What is the role of Functional Modeling in software design?	II	K2	CO2	PO3
37.	What is an Activity Diagram in UML?	II	K2	CO2	PO3
38.	List out the primary purpose of a State Chart Diagram in UML?	II	K2	CO3	PO5
39.	List out the components of a Data Flow Diagram (DFD) ?	II	K2	CO4	PO3
40.	How does Functional Modeling contribute to system understanding?	II	K2	CO2	PO4
41.	Define requirement analysis.	III	K1	CO2	PO2
42.	What are the main objectives of requirement analysis?	III	K1	CO2	PO2
43.	List out the types of requirements in software engineering.	III	K4	CO2	PO1
44.	What is a functional requirement?	III	K1	CO2	PO1
45.	What is a non-functional requirement? Provide an example.	III	K1	CO2	PO1
46.	List the steps involved in the requirements analysis process.	III	K4	CO3	PO2
47.	List any three techniques for gathering software requirements.	III	K4	CO3	PO2
48.	What is the purpose of stakeholder analysis in requirement gathering?	III	K1	CO3	PO1
49.	Differentiate between interviews and questionnaires in requirement gathering.	III	K2	CO2	PO2
50.	What is the role of the <i>Observer</i> pattern?	III	K1	CO2	PO2
51.	What is an architectural style in software design?	III	K1	CO5	PO4
52.	Define the layered architecture style.	III	K1	CO5	PO12
53.	What are the advantages of a client-server architecture?	III	K1	CO5	PO12
54.	What is user interface (UI) design in software development?	III	K1	CO4	PO4
55.	List three principles of good UI design.	III	K4	CO4	PO12
56.	What is the purpose of usability testing in UI design?	III	K1	CO4	PO2

57.	Differentiate between UI and UX design.	III	K1	CO2	PO2
58.	Why is consistency important in user interface design?	III	K1	CO2	PO3
59.	What is user interface (UI) design in software development?	III	K1	CO4	PO3
60.	What is the role of a case study in understanding software design?	III	K1	CO3	PO12
61.	Differentiate between sequence diagrams and collaboration diagrams.	IV	K4	CO3	PO2
62.	Define software testing.	IV	K1	CO2	PO1
63.	What are the primary goals of software testing?	IV	K1	CO2	PO1
64.	Differentiate between verification and validation in testing.	IV	K4	CO3	PO2
65.	What is unit testing?	IV	K1	CO6	PO1
66.	Who typically performs unit testing?	IV	K1	CO6	PO1
67.	List two benefits of unit testing.	IV	K4	CO6	PO2
68.	What is black box testing?	IV	K1	CO6	PO2
69.	Mention two techniques used in black box testing.	IV	K1	CO6	PO2
70.	How does black box testing differ from white box testing?	IV	K3	CO6	PO10
71.	What is white box testing?	IV	K1	CO6	PO2
72.	List any two techniques used in white box testing.	IV	K4	CO6	PO2
73.	Define integration testing.	IV	K1	CO6	PO2
74.	What is regression testing?	IV	K1	CO6	PO2
75.	List any two scenarios where regression testing is performed.	IV	K4	CO4	PO3
76.	What is system testing?	IV	K1	CO6	PO1
77.	List any three common debugging techniques.	IV	K4	CO1	PO3
78.	Compare testing and debugging.	IV	K2	CO1	PO2
79.	What is program analysis?	IV	K1	CO1	PO2
80.	Differentiate between static analysis and dynamic analysis.	IV	K4	CO1	PO2
81.	Define software project management.	V	K1	CO2	PO10
82.	What are the main objectives of project management?	V	K1	CO2	PO10
83.	List the stages of the project management lifecycle.	V	K4	CO2	PO10
84.	What is software configuration management?	V	K1	CO2	PO10
85.	List the primary components of SCM.	V	K4	CO1	PO10
86.	Define version control in SCM.	V	K1	CO2	PO2
87.	What is project scheduling?	V	K1	CO2	PO2
88.	Define a Gantt chart and its purpose.	V	K1	CO2	PO2
89.	List any two benefits of using project scheduling.	V	K4	CO1	PO2
90.	What is the critical path in project scheduling?	V	K1	CO1	PO2
91.	Differentiate between effort estimation and schedule estimation.	V	K1	CO1	PO2
92.	What is DevOps?	V	K1	CO7	PO1
93.	List any two key principles of DevOps.	V	K4	CO7	PO2
94.	What is cloud computing?	V	K1	CO8	PO2

95.	List two advantages of using the cloud as a platform in DevOps.	V	K4	CO7	PO3
96.	Mention three types of cloud service models.	V	K3	CO8	PO3
97.	List the main stages of a deployment pipeline.	V		CO1	PO2
98.	What is the significance of automated testing in the deployment pipeline?	V	K3	CO2	PO2
99.	Mention two tools used for configuration management in DevOps.	V	K4	CO7	PO3
100.	What is the role of Docker in DevOps?	V	K1	CO7	PO3

SECTION – B (8 MARKS)

S.No	Questions	Unit	K Level	CO	PO
101.	Explain the importance of Software Engineering and its key objectives in the context of software development.	I	K2	CO2	PO4
102.	Describe the Software Development Life Cycle (SDLC) and its different phases	I	K3	CO3	PO3
103.	Discuss the role of Software Engineering in ensuring quality, scalability, and maintainability of software systems	I	K3	CO4	PO2
104.	Explain the concept of a Software Process and describe its significance in software development	I	K2	CO2	PO3
105.	Discuss the various types of Software Process Models.	I	K2	CO3	PO4
106.	Analyze the V-Model and explain its advantages and disadvantages in software testing and development.	I	K3	CO4	PO3
107.	Discuss the role and importance of Specialized Process Models, such as the Spiral and Agile models	I	K3	CO3	PO3
108.	Describe the Agile Process and its various phases. How does it support flexibility and responsiveness in software development?	I	K3	CO4	PO5
109.	Explain Extreme Programming (XP) and its core practices	I	K3	CO3	PO3
110.	How does XP ensure high-quality software development?	I	K3	CO4	PO2
111.	How do Agile and XP methodologies ensure that the final product aligns with customer needs?	I	K2	CO4	PO3
112.	Discuss the role of customer feedback in Agile and XP	I	K2	CO3	PO2
113.	Explain the key practices of Extreme Programming (XP) and discuss how they contribute to delivering high-quality software	I	K3	CO4	PO3
114.	Explain the importance of "Test-Driven Development" (TDD) in XP and its impact on software quality	I	K2	CO2	PO2
115.	How does Extreme Programming (XP) ensure that the software is flexible and maintainable over time?	I	K3	CO3	PO2
116.	What is the role of the "Customer" in the XP process?	I	K2	CO4	PO3
117.	Explain the process of Requirement Gathering and Analysis in software engineering.	II	K3	CO2	PO3

118.	Discuss the structure and components of a Software Requirement Specification (SRS) document	II	K2	CO2	PO3
119.	Discuss the benefits and challenges of using formal methods in system specification	II	K2	CO3	PO3
120.	Compare and contrast Formal and Informal System Specifications.	II	K2	CO2	PO3
121.	Explain the concept of Finite State Machines (FSM) and how they are used in modeling system behavior	II	K4	CO2	PO3
122.	What are Petri Nets, and how can they be used for modeling concurrent systems?	II	K2	CO3	PO3
123.	Discuss their advantages and challenges in system modeling.	II	K2	CO2	PO3
124.	What are Interaction Diagrams in UML? Explain the difference between Sequence Diagrams and Communication Diagrams	II	K2	CO2	PO3
125.	Explain the role and purpose of an Activity Diagram in UML.	II	K2	CO2	PO3
126.	Describe a State Chart Diagram in UML.	II	K2	CO2	PO3
127.	Explain Data Flow Diagrams (DFD). How do DFDs help in analyzing system processes and their interactions?	II	K2	CO3	PO2
128.	Discuss the different levels of DFDs with examples.	II	K2	CO3	PO3
129.	Explain its significance in the context of system design.	II	K2	CO2	PO2
130.	Explain an activity diagram for an online shopping system.	II	K2	CO3	PO3
131.	Explain an activity of a class diagram for a simple library management system.	II	K2	CO2	PO3
132.	Explain the significance of State Chart Diagrams in object-oriented design?	II	K2	CO2	PO3
133.	Write short notes on design process in software development.	III	K6	CO1	PO10
134.	Define design concepts in software engineering.	III	K1	CO1	PO10
135.	Differentiate between coupling and cohesion in software design. Provide examples of low coupling and high cohesion.	III	K2	CO1	PO10
136.	What is functional independence ? Explain with examples.	III	K1	CO3	PO10
137.	Explain the publish-subscribe design pattern.	III	K2	CO3	PO10
138.	Explain about the Adapter , Command , and Strategy design patterns	III	K2	CO3	PO10
139.	Summarize the advantages and applications of design patterns.		K3	CO2	PO10
140.	Compare the Layered Architecture and Client-Server Architecture . Discuss their pros and cons.	III	K3	CO5	PO5
141.	Explain the Pipe and Filter Architecture with an example.	III	K2	CO5	PO5
142.	What are the main features of a Tiered Architecture ?	III	K1	CO5	PO5
143.	Explain with examples of good and poor UI designs.	III	K2	CO2	PO5
144.	What are the challenges faced in user interface design , and how can they be addressed?	III	K1	CO2	PO5

145.	Explain the Model-View-Controller (MVC) design pattern with an example of its application.	III	K2	CO2	PO5
146.	Explain the publish-subscribe design pattern. Discuss its advantages and applications.	III	K2	CO2	PO8
147.	Analyze a case study where the MVC pattern was applied effectively. Highlight its impact on design and maintenance.	III	K4	CO3	PO8
148.	Show a case study of a software system using the Facade pattern to manage system complexity.	III	K1	CO2	PO8
149.	Define software testing and explain in detail.	IV	K1	CO2	PO1
150.	Discuss about examples of tools used for unit testing	IV	K1	CO7	PO3
151.	Summarize the Difference between black box testing and white box testing with examples.	IV	K2	CO7	PO2
152.	Explain the advantages and disadvantages of black box and white box testing.	IV	K2	CO7	PO2
153.	What is integration testing ? Illustrate with examples.	IV	K1	CO7	PO2
154.	Discuss different approaches to integration testing.	IV	K1	CO7	PO1
155.	What is regression testing , and why is it critical in software maintenance?	IV	K1	CO7	PO3
156.	How is automation can improve regression testing and list tools used for this purpose.	IV	K3	CO7	PO2
157.	Define debugging and explain the typical process followed to identify and fix defects in software.	IV	K1	CO2	PO2
158.	What are the challenges associated with debugging and strategies to overcome them.	IV	K1	CO2	PO2
159.	What is program analysis ?	IV	K1	CO1	PO1
160.	Explain how program analysis helps in improving software quality.	IV	K1	CO2	PO3
161.	Define symbolic execution and explain how it is used to analyze programs.	IV	K1	CO1	PO2
162.	What are the limitations of symbolic execution, and how are they addressed in modern tools?	IV	K1	CO2	PO2
163.	What is model checking ? Discuss its importance in verifying software and hardware systems.	IV	K1	CO2	PO2
164.	Explain with example of how model checking is applied to ensure the correctness of a system.	IV	K1	CO2	PO4
165.	Explain the importance of Software Configuration Management (SCM) and in software projects.	V	K1	CO1	PO5
166.	What are version control systems?	V	K1	CO1	PO5
167.	Describe the agile projects and suggest strategies to address them.	V	K1	CO2	PO5
168.	Describe the critical path method (CPM) and its application in software project scheduling. Provide an example.	V	K1	CO1	PO5
169.	What are the key steps involved in creating a software project schedule?	V	K1	CO1	PO5
170.	Explain the role of Gantt charts and PERT charts in project scheduling. Highlight their advantages and limitations.	V	K1	CO1	PO5

171.	Explain the core principles of DevOps and discuss its role in bridging the gap between development and operations teams.	V	K1	CO7	PO4
172.	Discuss the key advantages of adopting DevOps practices in software projects.	V	K1	CO7	PO4
173.	What are the key advantages of using cloud platforms for software development and deployment	V	K1	CO7	PO4
174.	Compare and contrast Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS), and Software-as-a-Service (SaaS) in cloud computing.	V	K1	CO7	PO4
175.	What is a deployment pipeline? Explain its stages and the importance of automation in this process.	V	K1	CO1	PO4
176.	Discuss the concept of Continuous Integration and Continuous Delivery (CI/CD) in the context of deployment pipelines.	V	K1	CO2	PO4
177.	Discuss the types of automated testing	V	K1	CO2	PO4
178.	Describe the process of building software in a CI/CD pipeline	V	K1	CO1	PO4
179.	What are the various deployment strategies (e.g., Blue-Green, Canary, Rolling) used in software projects?	V	K1	CO2	PO2
180.	Explain how containerization (e.g., Docker) simplifies deployment. Provide an example of its use in a software project.	V	K1	CO2	PO2

SECTION – C (15 MARKS)

S. No	Questions	Unit	K Level	CO	PO
181.	Compare and contrast the Waterfall Model, Incremental Model, and Agile Process Model in terms of their advantages and disadvantages.	I	K4	CO3	PO2
182.	Discuss the various perspectives of software engineering, including the technical, managerial, and economic perspectives. How do these perspectives influence the software development process?	I	K4	CO2	PO3
183.	Compare and contrast the Agile and Waterfall software development methodologies. Discuss the situations where each model is most effective	I	K3	CO4	PO3
184.	Explain the concept of the "Spiral Model" of software development.	I	K3	CO2	PO2
185.	Explain the Agile software development process in detail.	I	K3	CO3	PO3
186.	What is Extreme Programming (XP)? Explain its core values, practices, and how it enhances software quality.	I	K5	CO2	PO2
187.	Compare and contrast Agile and Extreme Programming (XP).	I	K5	CO4	PO3
188.	Explain how Continuous Integration (CI) works in Agile and XP	I	K3	CO4	PO3
189.	Discuss various techniques used for requirements gathering and the challenges involved in this phase	II	K3	CO2	PO2

190.	Discuss its structure and components, and explain the importance of an SRS in software development.	II	K3	CO2	PO3
191.	How is it different from informal requirements documentation, and what are its advantages in the software development process?	II	K3	CO3	PO2
192.	Discuss the importance of Requirements Analysis and Specification in the software development lifecycle	II	K3	CO3	PO4
193.	Explain the concept of Finite State Machines (FSM). Discuss the components of FSMs, how they are used in system modeling, and their advantages and limitations	II	K3	CO2	PO2
194.	Explain the use of Use Case Models and Class Diagrams in UML with examples.	II	K3	CO3	PO4
195.	Explain the concept of Activity Diagrams in UML. Discuss their role in modeling workflows and business processes.	II	K3	CO2	PO3
196.	Discuss the role of CASE (Computer-Aided Software Engineering) tools in software development	II	K3	CO3	PO4
197.	Explain the design process in software development. Discuss the key stages with examples.	III	K1	CO1	PO4
198.	Elaborate on abstraction, modularity, and information hiding .	III	K1	CO2	PO4
199.	What is functional independence , and why is it significant in software design? Explain with examples.	III	K1	CO1	PO4
200.	Write short notes on the Adapter, Command, and Strategy design patterns with use-case scenarios.	III	K1	CO2	PO4
201.	Discuss the key principles of user interface design .	III	K2	CO1	PO4
202.	Explain the Model-View-Controller (MVC) design pattern. Discuss its components and provide an example of its application.	III	K2	CO4	PO4
203.	Describe the Publish-Subscribe pattern. How does it differ from Observer, and what are its advantages in real-world applications?	III	K2	CO44	PO4
204.	What is the Adapter pattern? Explain its purpose with a diagram and an example.	III	K2	CO4	PO4
205.	Differentiate between static analysis and dynamic analysis with examples.	IV	K2	CO4	PO4
206.	Explain in detail about software development lifecycle.	IV	K2	CO1	PO1
207.	Discuss the importance of in the software testing in development lifecycle.	IV	K1	CO6	PO1
208.	Compare and contrast black box testing and white box testing with examples of when each approach is appropriate.	IV	K1	CO6	PO4
209.	What is integration testing , and how does it differ from system testing ? Illustrate with examples.	IV	K1	CO6	PO4
210.	Define symbolic execution and Provide examples of scenarios where it is beneficial.	IV	K1	CO4	PO4
211.	Describe unit testing and discuss its role in ensuring software quality	IV	K1	CO6	PO4
212.	Explain in detail about top-down, bottom-up, and sandwich testing.	IV	K2	CO6	PO4

213.	Explain the key activities involved in Software Configuration Management (SCM) and discuss its importance in software projects.	V	K2	CO2	PO5
214.	Discuss the challenges of managing software configuration in agile projects and suggest strategies to them.	V	K2	CO6	PO5
215.	Explain the critical path method (CPM) and its application in software project scheduling. Provide an example.	V	K2	CO2	PO5
216.	What are the key steps involved in creating a software project schedule?	V	K2	CO2	PO5
217.	What are the key advantages of using cloud platforms for software development and deployment?	V	K1	CO8	PO5
218.	Discuss the key benefits of adopting DevOps practices in software projects. Include examples to support your answer.	V	K2	CO7	PO5
219.	Explain the significance of automated testing in the DevOps pipeline.	V	K2	CO6	PO12
220.	Compare centralized and distributed version control systems with examples.	V	K2	CO7	PO12

K1- Remember; K2- Understand; K3- Apply; K4- Analyze;

K5-Evaluate; K6-Create